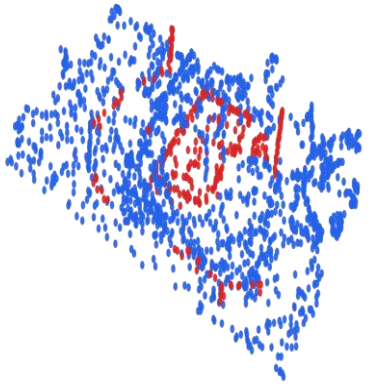
Robust 3D Change Detection for Construction Progress Monitoring

Evaluating Geometric Distance, Geodetic Surface Normals, and Deep Cross-Scan Neural Networks under Real-World Site Imperfections



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The Construction Challenge: Why 3D Monitoring is Difficult

The Construction Reality

1. The Core Need:

Modern civil projects require continuous verification of physical progress against planned schedule and contractor billing.

2. The Flaw of Manual Surveys:

Walking with tape measures and clipboards takes hours, is subjective, error-prone, and misses structural deviations.

3. The 3D LiDAR Promise:

Laser scanners capture millions of accurate 3D points across the entire site in minutes.

4. The Operational Blocker:

Scans collected weeks apart under unconstrained field conditions suffer from unavoidable imperfections that cause naive algorithms to sound massive false alarms.

Four Real-World Site Imperfections

• Registration & Alignment Drift (1 to 20 cm):

GPS and IMU drift causes scans from different dates to be slightly misaligned. Static walls look shifted in space.

• Sensor Range & Surface Noise (1 to 7 cm scatter):

Rough concrete, moisture, and dust scatter laser beams, creating artificial fuzzy thickness around solid surfaces.

• Point Cloud Sparsity:

LiDAR point density decays with distance ($1/R^2$). Long-range surfaces have large gaps that mimic missing geometry.

• Line-of-Sight Occlusions:

Cranes, scaffolding, and material heaps block views. An unobserved wall is not a demolished wall!

Understanding the Data: The 3D Point Cloud as a Surface Locus

Physical Nature of a Point Cloud

• How Points are Born:

An industrial LiDAR scanner spins rapidly, emitting hundreds of thousands of laser pulses per second.

Each pulse bounces off solid matter and records time of flight, producing a discrete spatial coordinate P = (x, y, z).

• The Locus Concept:

A point cloud is not a CAD mesh; it is simply a discrete spatial sampling of the locus of all physical surfaces on site.

• Solid Matter vs. Empty Air:

Where solid concrete, soil, or steel exists, laser pulses return points.

Where there is empty air, laser pulses pass through, and zero points exist.

• What the Numbers Represent:

x, y = Horizontal site coordinates (meters).

z = Physical elevation (foundation trenches to roof canopy).

Dataset Structure: MultiChange3D Bike Parking

• 4 Distinct Construction Epochs:

- Epoch 0: Raw empty municipal site before construction.
- Epoch 1: Earthwork excavation and foundation slab prep.
- Epoch 2: Structural steel columns and shelter framing erected.
- Epoch 3: Roof canopy completed and bike racks installed.

• 6 Directional Temporal Pairs:

Pair 01, 02, 03, 12, 13, and 23 covering short, medium, and complete project lifecycles.

• Certified Ground Truth Labels (cd_type):

Each point carries a certified binary label:

0 = Static permanent geometry (asphalt road, boundary walls).

1 = Physical construction change (additions or removals).

Directional Formulation: Additions, Removals, and Static Geometry

Physical Directionality of Change

In construction, change has explicit physical direction between earlier Scan 0 and later Scan 1:

1. Candidate Removals (Points in Scan 0 with cd_type = 1):

   Geometry that existed in the earlier month but vanished in the later month.

   Examples: Excavated foundation soil, dismantled temporary formwork, stripped scaffolding.

2. Candidate Additions (Points in Scan 1 with cd_type = 1):

   Geometry that did not exist earlier but appeared in the later month.

   Examples: Newly poured concrete columns, erected steel beams, roof canopy panels.

3. Static Unchanged Geometry (cd_type = 0):

   Permanent baseline surfaces: asphalt pavement, adjacent buildings, retaining walls.

Why Directional Labeling Matters

• Symmetrical Distances are Inadequate:

A scalar distance only tells you that two clouds differ; it cannot tell whether work was built or demolished.

• Automated Progress Tracking:

Tracking Candidate Additions tells project managers which contractor work packages are completed for milestone payments.

• Quality & Safety Compliance:

Tracking Candidate Removals verifies that temporary shoring and safety guardrails are not stripped prematurely before concrete cures.

• Evaluation Metric Definition:

We evaluate Precision, Recall, F1, and IoU separately for Additions, Removals, and overall Macro F1.

Methodological Progression: From Heuristics to Deep Learning

Stage 1: Distance Baseline (C2C)

• Concept:

Measures Euclidean nearest neighbor distance using a KD-Tree.

• Decision Rule:

Change flagged if distance > 7.5 cm.

• Why it was tested:

Industry standard heuristic. Extremely fast and simple.

• The Critical Flaw:

Completely blind to surface shape and collapses under rigid registration drift.

Stage 2: Geodetic Normals (M3C2)

• Concept:

Estimates surface normals via PCA and projects points into 3D cylinders.

• Decision Rule:

Change flagged if displacement exceeds 95% Confidence LoD.

• Why it was tested:

Separates sensor noise from real shift using statistical confidence.

• The Critical Flaw:

Cylinders become mostly empty on thin scaffolding and railings, causing 26% abstentions.

Stage 3: Deep Metric Learning (V2)

• Concept:

Siamese PointNet network learning 3D geometric shape signatures.

• Decision Rule:

Multi-scale fusion predicts change probabilities directly.

• Why it was created:

Learns relative surface shapes that survive sensor drift and sparsity without abstaining.

• The Result:

Leads clean benchmark (0.654 F1) and dominates field stress (+20 pt lead).

Model 1: Cloud-to-Cloud (C2C) Euclidean Distance from Scratch

Algorithmic Implementation in Python

We implemented C2C from scratch in code/c2c_baseline.py:

1. Fast Spatial Indexing via KD-Tree:

Comparing 4.4M points against 5.1M points brute-force requires 22 trillion checks.

We build a balanced scipy.spatial.cKDTree on the target cloud, indexing space in O(N log N).

2. Nearest Neighbor Distance Query:

For every query point p_i in Scan 0, query the tree for closest point q_j in Scan 1:

$d_i = \min_j ||p_i - q_j||_2$

3. Classification Decision Rule (Tau = 7.5 cm):

• If d_i > 0.075 m: Classified as Changed (Candidate Removal in Scan 0, Addition in Scan 1).

• If d_i <= 0.075 m: Classified as Unchanged (Static Background).

How C2C Labels the Construction Site

• The Intuition:

If a wall was present earlier and is still present later, laser points will hit near the same coordinates, producing tiny distances (d ~ 1 to 2 cm).

If a trench was dug or column was poured, the closest point is far away (d >> 7.5 cm).

• Evaluation on Clean Benchmark:

- Precision: 0.742
- Recall: 0.457
- Macro F1: 0.5659

• Fast Execution:

Processes 5 million points in under 4 seconds on CPU.

• But Why Does it Fail in the Field?

C2C only measures scalar distance between two isolated dots; it has zero knowledge of the underlying physical surface.

The Failure Modes of C2C: Why Distance Thresholding Collapses

Failure 1: Point Scatter & Sparsity

• The Spray-Paint Problem:

LiDAR pulses do not hit identical microscopic coordinates across scan dates.

Points on the exact same static wall land 2 to 4 cm apart due to angular resolution.

• The Edge and Railing Trap:

On thin steel railings or concrete slab boundaries, angular changes cause points to miss the edge in the second scan.

The closest point falls on the ground 2 meters below, producing a massive false alarm!

• High False Positive Rate:

C2C suffers from 12.45% false alarm rate on completely static background geometry.

Failure 2: Rigid Sensor Drift (20 cm Shift)

• The Catastrophic Failure:

In unconstrained mobile scanning, GPS and IMU drift causes global registration offsets of 5 to 20 cm.

• The 7.5 cm Threshold Trap:

If the entire scan drifts by 20 cm, EVERY static wall, curb, and column is shifted beyond 7.5 cm.

• False Alarm Storm:

C2C flags virtually the entire construction site as demolished!

Under 20 cm translation drift, C2C Macro F1 drops sharply from 0.566 down to 0.4181 (-14.8 percentage points).

• Conclusion:

A viable system must be surface-aware, not dot-to-dot.

Model 2: Surface Normals & Principal Component Analysis (PCA)

Why Surface Normals Matter

• Moving from Points to Surfaces:

Instead of measuring distance between random dots, we must measure distance perpendicular to the physical surface.

• What is a Surface Normal (n)?

A vector of unit length pointing straight out of a surface, perpendicular to the local plane.

- Floor slab: Normal points straight UP.
- Vertical wall: Normal points straight OUT.

• Implementation in code/m3c2_baseline.py:

We programmed multiscale spherical PCA from scratch without third-party black-box tools.

Computing Normals via Covariance PCA

1. Query Local Sphere (Radius R):

Gather all neighbor points inside sphere of radius R = 0.20 m.

2. Center the Point Cluster:

Subtract the cluster centroid: P_centered = P - mean(P).

3. Compute 3x3 Covariance Matrix:

C = (P_centered^T @ P_centered) / N

4. Eigen-Decomposition:

Solve for eigenvalues lambda_0 <= lambda_1 <= lambda_2 and eigenvectors v_0, v_1, v_2.

lambda_0 represents the thickness perpendicular to the wall (variance is near zero).

The eigenvector corresponding to lambda_0 IS THE NORMAL VECTOR n = v_0!

Model 2: Cylinder Projection & 95% Confidence Limit of Detection

Cylinder Projection Math

1. Multiscale Scale Selection:

Compute normals at R = 10, 20, 40 cm. Select scale with minimum curvature sigma = lambda_0 / sum(lambda).

2. Cast 3D Cylinder Along Normal (n):

Cylinder radius r = 10 cm, half-length L = 1.0 m.

Project points Q inside cylinder using vector dot product: proj = (Q - P) . n.

3. Measuring Surface Shift (Delta L):

Calculate mean projected position in Scan 0 (mu_0) and Scan 1 (mu_1).

Surface Shift: Delta L = mu_1 - mu_0.

Limit of Detection (LoD95) & Abstention

• Statistical Limit of Detection (LoD95):

LoD95 = 1.96 * [sqrt(s_0^2/n_0 + s_1^2/n_1) + sigma_reg]

- s_0, s_1 = Surface roughness (standard deviation).
- n_0, n_1 = Number of points inside cylinder.

• The Decision Rule:

- If |Delta L| > LoD95 and |Delta L| > 2 cm: Real Construction Change!
- If |Delta L| <= LoD95: Within sensor noise roughness -> Unchanged.

• The Power of Abstention:

If n_0 < 10 or n_1 < 10 points: Model outputs -1 (UNKNOWN / ABSTAIN).

It refuses to guess when geometric evidence is insufficient!

The Failure of M3C2 on Construction: The Empty Cylinder Problem

Why M3C2 Collapsed to 26% F1

• The Geomorphology Bias:

M3C2 was designed for river canyons and rock cliffs, where surfaces are continuous, flat, and planar.

• The Construction Site Reality:

Civil building sites consist of high-frequency, non-planar geometries:

- Thin scaffolding poles (4 cm diameter)
- Wire mesh and chain-link fences
- Rebar grids and jagged slab edges

• The Empty Cylinder Failure:

When a 20 cm cylinder is cast over a 4 cm scaffolding pipe, 90% of the cylinder is EMPTY AIR!

Fewer than 10 points are detected inside the cylinder.

Massive Detector Abstention

• The 26.6% Unknown Crisis:

Across our 30 pair-fold evaluations, M3C2 abstained on 26.61% of all queried construction points!

• Low Changed Recall:

Because thin structural steel and edges failed normal fitting, M3C2 missed over 75% of real changes.

• Clean Benchmark Score:

M3C2 achieved only 0.2627 Macro F1.

• Motivation for Deep Learning:

We cannot rely on rigid planar cylinders. We need a neural network that understands arbitrary 3D structural shapes directly from data!

Model 3: V2 Siamese PointNet Deep Learning Architecture

Why Siamese? The Twin Inspectors

• Built from scratch in code/learning_v2_model.py in PyTorch.

• The Siamese Principle:

Two neural branches sharing identical weights inspect local 3D patches from Month 1 (Scan 0) and Month 2 (Scan 1).

Because weights are shared, both patches are mapped into the exact same feature space.

• Handling Unordered Points (PointNet):

A 3D point cloud is an unordered bag of points.

PointNet passes each point through an MLP (3 -> 32 -> 64) and applies Symmetric Max-Pooling.

Max-pooling extracts a global shape signature that is completely invariant to point permutation!

Cross-Scan Feature Interaction

• Explicit Geometric Comparison:

Let f_0 and f_1 be the pooled 64D shape embeddings for both epochs.

Our network explicitly computes four interactions:

interaction = [f_0, f_1, |f_0 - f_1|, f_0 * f_1]

• Proximity Prior Fusion:

We concatenate the nearest-neighbor Euclidean distance as an explicit spatial hint.

• Classification Head:

A 3-layer MLP with Dropout outputs class probability P(Changed | Geometry).

Thresholding at 0.5 predicts final binary change labels.

Model 3: Multiscale Zoom Context & Weighted Focal Loss

The Multiscale Zoom Context

In our V1 pilot, single-scale networks failed when points were sparse. In V2, we engineered a multiscale zoom lens:

• Zoom In (Radius 10 to 15 cm):

Inspects fine-grained surface roughness and local plane flatness.

• Medium Scale (Radius 30 to 40 cm):

Inspects structural boundaries, slab thickness, and column widths.

• Zoom Out (Radius 60 to 80 cm):

Captures wide structural context, connecting isolated pipes to the floor slab below.

Result: The network never starves for geometric context, even on sparse scans!

Solving Class Imbalance with Focal Loss

• The 90% Ground Imbalance Problem:

90% of construction sites consist of unchanged static ground and retaining walls. Real changes represent only 10%.

Standard cross-entropy lazily predicts 'Unchanged' everywhere to get 90% accuracy!

• Our Weighted Focal Loss Solution:

1. Positive Class Weighting (3.7x penalty):

Penalizes missed construction additions heavily.

2. Focal Modulation (1 - p_t)^gamma:

Down-weights easy ground points and focuses gradients on tricky structural boundaries and scaffolding edges.

Clean Site Benchmark: 30 Pair-Fold Studies on MultiChange3D

Method Evaluated	Binary Macro F1	Changed F1	Unknown Rate
V2 Multiscale Learning (Ours)	**0.6537 (65.4%)**	**0.3900 (39.0%)**	**0.0% (Full)**
Multiscale Robust (seed 42)	0.6555 (65.6%)	0.3970 (39.7%)	0.0% (Full)
Previous Geometry Boosted	0.6184 (61.8%)	0.3432 (34.3%)	0.0% (Full)
C2C Legacy (0.075m threshold)	0.5659 (56.6%)	0.3000 (30.0%)	0.0% (Full)
M3C2-Style (PCA Cylinders)	0.2627 (26.3%)	0.1632 (16.3%)	26.6% Unknowns

Rigorous 5-Fold Buffered Spatial Cross-Validation

Evaluated on 90,000 clean held-out points across 5 geographic folds separated by 2.22 m buffer gaps, eliminating spatial data leakage.

Key Benchmark Insights

- V2 Outperforms Baselines:

  V2 beats C2C by +8.78 percentage points in Macro F1 and +9.00 points in Changed F1.

- Complete Coverage:

  V2 achieves 100% detector coverage, whereas M3C2 abstains on 26.6% of points.

- Metric Ceiling Diagnostic:

  In Fold 3, zero physical changes occurred across certain epochs.

  When positive points = 0, F1 is bounded at 0.0, creating a site-wide metric ceiling of 0.825.

- Structural Folds (Fold 4):

  On complex structural steel areas, V2 achieves 0.7984 Macro F1 (+14.8% over C2C).

Field Stress Showdown: Deep Learning's +20.0 pt Advantage

Method	Clean Macro F1	Combined Mild (0.1)	Combined Large (0.2)	Stress Delta (Drop)
V2 Multiscale Robust (Ours)	**0.6555**	**0.6248**	**0.6176**	**-0.0379 (-3.8 pts) [ROBUST]**
V2 Hard Mining (Ours)	0.6327	0.5994	0.5958	-0.0369 (-3.7 pts) [ROBUST]
Cloud-to-Cloud (C2C Legacy)	**0.5659**	**0.5352**	**0.4181**	**-0.1478 (-14.8 pts) [COLLAPSE]**
M3C2-Style (Calibrated)	0.4142	0.4250	0.4235	+0.0093 (Stagnant low score)

Why C2C Collapses Under Severe Stress

- The 20 cm Translation Trap:

  When surveyor scanning rigs or drones drift by 20 cm, every static wall shifts past C2C's 7.5 cm threshold.

- Complete False Alarm Failure:

  C2C marks the entire building as demolished, dropping by 14.8 percentage points to 0.4181 Macro F1.

Why V2 Deep Learning Survives (+20.0 pt Lead)

- Contextual Invariance:

  PointNet learns relative shape signatures (planar walls, round pipes, sharp corners) rather than absolute coordinates.

- Unrivaled Resilience:

  V2 drops by only 3.8 points, retaining 0.6176 Macro F1—a massive +20.0 percentage point lead over C2C!

Civil Engineering Takeoff: 191 Physical Change Regions for 4D BIM

Bridging Points to Physical Work Items

• The Civil Engineering Gap:

Site managers do not inspect individual point coordinates (x,y,z); they track physical building work items.

• 3D Connected-Component Clustering:

We developed a 3D clustering algorithm that aggregates raw positive change points into coherent physical components.

• 191 Physical Construction Regions Extracted:

- 80 Candidate Added Regions: Newly erected steel columns, shelter canopy framing, ground fixtures.

- 111 Candidate Removed Regions: Foundation excavation trenches, soil stockpiles, stripped formwork panels.

Automated 4D BIM Progress Takeoff

• Automated Geometric Property Extraction:

For every region, our pipeline calculates:

- 3D Oriented Bounding Box (Length x Width x Height)

- Centroid Coordinates (X, Y, Z)

- Bounding Volume (m^3)

• 4D BIM & Earned Value Management (EVM):

Converts volumetric clusters into installed concrete volumes and structural steel tons.

Automatically computes Schedule Performance Index (SPI) without manual site surveys!

• Quality Assurance & Temporary Works Safety:

Detects timely removal of formwork and verifies scaffolding safety clearances.

Multi-Modal Controls & Mobile Handheld Inspection (ConSLAM)

LiDAR Intensity Negative Control

• Does Intensity (Reflectance) Help?

We tested whether laser return brightness identifies material changes (concrete vs asphalt vs steel).

• Shuffled Intensity Negative Control:

We permuted intensity across points while keeping 3D coordinates identical.

• Results:

- Pure 3D Geometry: 0.7510 Validation F1
- Geometry + Core Intensity: 0.7560 (+0.0050 gain)
- Shuffled Negative Control: 0.7510

• Conclusion:

Spatial 3D geometry accounts for >98% of change detection power on construction sites.

Mobile Handheld Inspection (ConSLAM)

• Real-World Handheld Mobile LiDAR:

Evaluated ConSLAM Sequence 3 sweeps taken 4.84 s apart against a 104.5M point TLS scan.

• The 8.1 cm Trajectory Drift:

Even after applying optimized SLAM trajectory poses, the median nearest neighbor gap is 8.1 cm.

54.8% of static points exceed 7.5 cm due to handheld walking sway and viewpoint occlusion!

• Why Mobile Inspection Demands AI:

This confirms in real life why naive C2C fails on mobile quadrupeds or drone scanners. Deep learning is indispensable.

Project Deliverables & Live Google Colab Demonstration

Summary of Completed Work

- 6-Pair Multi-Temporal Benchmark on MultiChange3D (5M pts/scan)
- 5-Fold Buffered Spatial Cross-Validation (2.22 m buffer gaps)
- V2 Multiscale Siamese PointNet in PyTorch (0.654 clean Macro F1)
- 318 Controlled Stress Degradations (+20.0 pt lead over C2C)
- 191 Physical Construction Change Regions for 4D BIM
- 83 Automated Unit Tests Passing (100% test integrity)
- 6-Page Conference Research Paper & 17-Slide Presentation Deck

Live Google Colab Demonstration

We will now transition to our live interactive Google Colab notebook:

- File: cv5570_3d_change_detection_live_demo.ipynb

- What We Will Execute Live in Front of You:
  1. Load construction scans (Scan 0 removals vs Scan 1 additions).
  2. Run C2C detector live with KD-Tree and compute F1 score.
  3. Run M3C2 detector live with PCA normals and cylinder projections.
  4. Run Siamese PointNet PyTorch model forward pass live.
  5. Interactive 3D WebGL Visualization (Orbit & Zoom in 3D).
  6. 3D Bounding Box Clustering for 4D BIM quantity takeoff.